

$$S_{\text{sky}}: \frac{10000 \text{ V}}{10 \text{ s}} = \boxed{1000 \frac{\text{V}}{\text{s}} = R_{\text{sky}}} \leftarrow$$

$$S_{\text{sky} + \text{obj}}: \frac{5500 \text{ V}}{5 \text{ s}} = 1100 \frac{\text{V}}{\text{s}} = R_{\text{sky}} + R_{\text{source}} *$$

$$1100 \frac{\text{V}}{\text{s}} = 1000 \frac{\text{V}}{\text{s}} + R_{\text{source}}$$

$$\boxed{R_{\text{source}} = 100 \frac{\text{V}}{\text{s}}}$$

$$\sigma_{\text{sky}} = \frac{\sqrt{N_{\text{sky}}}}{t}$$

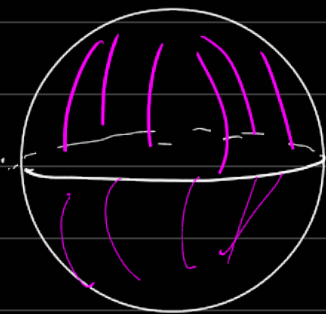
$$\frac{S}{N} \sim \frac{Nt}{\sqrt{Nst}} \leftarrow$$

$$\sigma_{\text{obj}} = \frac{\sqrt{N_{\text{obj}}}}{t}$$

(12) $\boxed{\frac{\Delta L}{L_0} = \alpha \Delta T}$ (linear expansion equation)

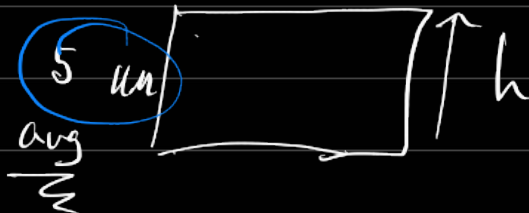
$$\frac{\Delta V}{V_0} = \alpha \Delta T \leftarrow \Delta V = V_0 \alpha \Delta T$$

(Vol of Ocean)



$$SA = 4\pi R_{\oplus}^2$$

$$A_{\text{ocean}}: \frac{7}{10} (4\pi R_{\oplus}^2) \quad SA$$



$$(Vol)_0 = \frac{35 \text{ km}}{10} (4\pi R_{\oplus}^2)$$

$$R_{\oplus} \sim 6300 \text{ km}$$

$$\Delta V = \Delta h \cdot (SA) = (SA \cdot h) \cdot \alpha \Delta T$$

$$\Delta h = h \alpha \Delta T$$

* expansion only occurs up north *

$$\Delta h = (\sum_n (1. \times \omega_n^{-4} k^4)) (2k)$$

$$= 1 \text{ m}$$

